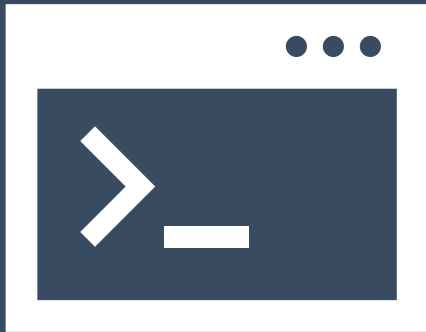


Introduction to Spatial Transcriptomics

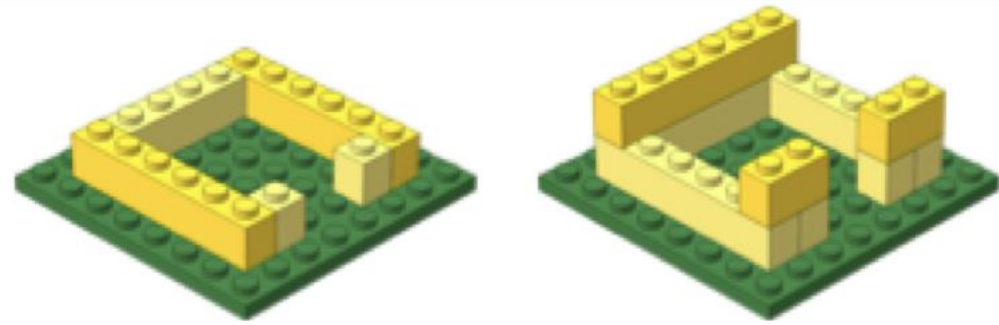
<https://tinyurl.com/hbc-spatial>



Harvard Chan Bioinformatics Core



Learning Objectives

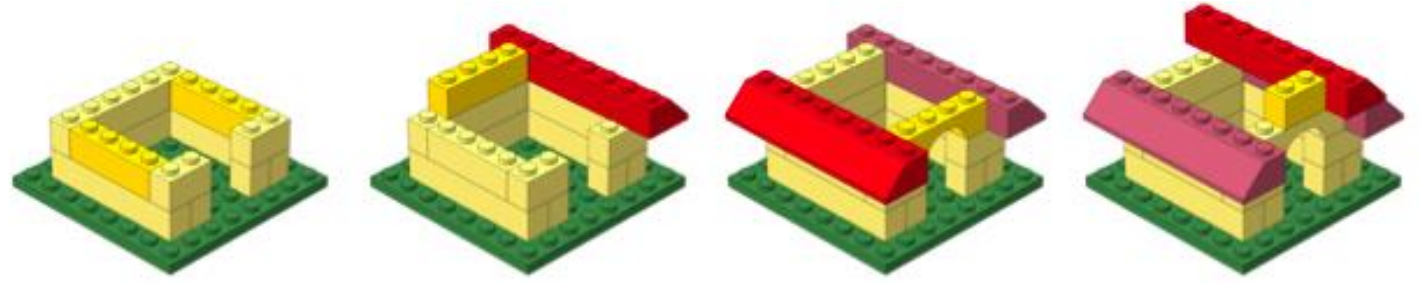


- ❖ Experimental design best practices of current spatial transcriptomics technologies
- ❖ Data loading and quality control of Visium HD samples
- ❖ Standard single-cell workflows adapted for spatial data, including highly variable gene selection, PCA, UMAP, kNN, and clustering
- ❖ Cell type annotation with deconvolution using RCTD for sequencing-based technologies
- ❖ Spatial-specific analyses that utilize the physical location of bins/cells on the tissue
 - ❖ Spatial clustering with BANKSY
 - ❖ Spatially variable gene detection with Moran's I
 - ❖ Cell-cell communication with CellChat

Exit survey

<https://tinyurl.com/hbc-spatial-survey>

Keep building!

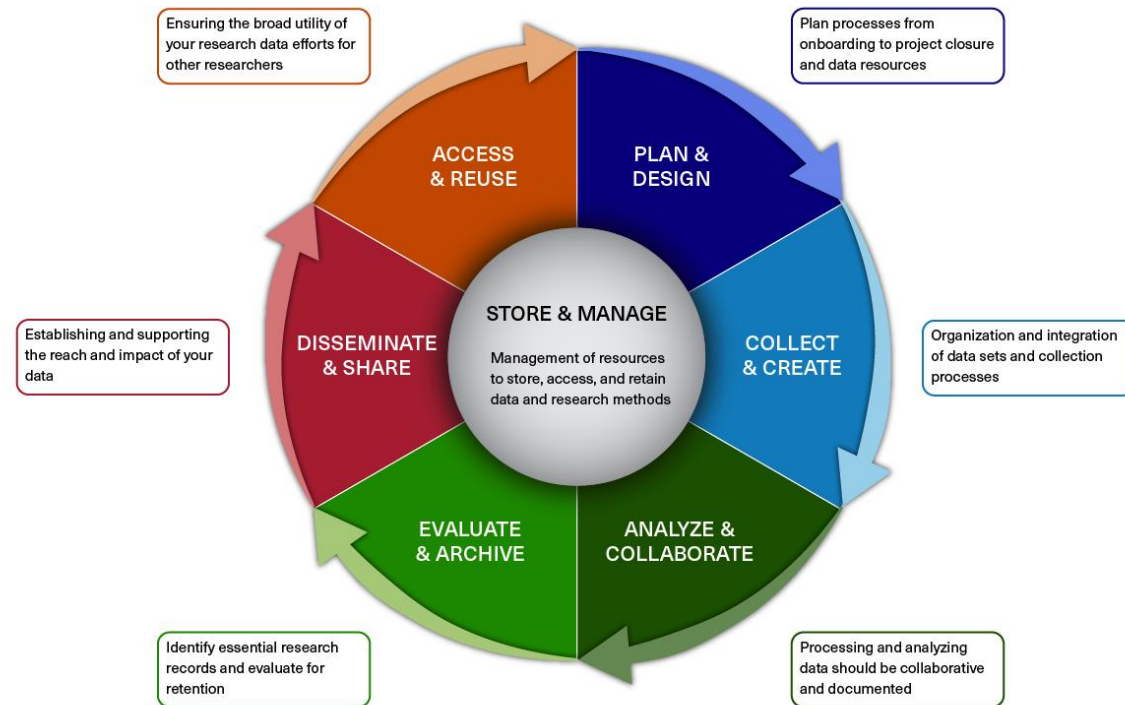


Module	Pre-requisite	Date	Time	Registration Page
Getting Started with Nextflow Workflow Development	Basic experience with command line and scripting concepts (e.g. bash, python, R or equivalent)	6/10/26	9:30am-12:30pm	Register here
Making Sense of Single-Cell Data: Practical Exploration and Interpretation with Loupe Browser	None	7/15/26	9:30am-12:30pm	Register here
Machine Learning for Biologists	None	8/19/26	9:30am-12:30pm	Register here
The cBioPortal for Cancer Genomics	None	9/23/26	9:30am-12:30pm	Register here
Single Cell RNA-seq Analysis Using Cellenics	HarvardKey Credentials	10/21/26	9:30am-12:30pm	Register here

<https://hsph.harvard.edu/research/bioinformatics/training/training-schedule/>

Research Data Management (RDM)

BIOMEDICAL RESEARCH DATA LIFECYCLE



Better RDM practice benefits you

❖ HMS Data Management LMA

❖ **Webpage:** <https://datamanagement.hms.harvard.edu>

❖ Sign up for quarterly email updates

❖ Harvard-wide Research data Management

❖ <https://researchdatamanagement.harvard.edu/>

Spring 2026 Data Lifecycle Training

Plan & Design

March 11 

The Lifecycle of
Scholarly 3D Data

April 8 

Data Management
with DMP Tool

May 6 

Offboarding for
Research Projects

Collect & Analyze

February 25 

Interact with RShiny

March 12 

Intro to MATLAB

March 26 

Public Health GIS

April 15 

Intro to Git & GitHub

April 29 

Prepare and Share Your
Software

April 30 

Tidy Up Your Data!

May 14 

Intro to Python

Store & Evaluate

February 26 

Finalize, Submit, and
Transfer Your Data

March 10 

Introduction to the
General Records Schedule

March 24 

Managing Your Paper
Records: Off-Site Records
Storage

April 21 

Managing Your Electronic
Records: Shared Drives
and Email

Share & Publish

February 5 

Share and Publish
Data with OSF

March 10 

The Making of a Data
Availability Statement

March 18 

Publication Perfect

May 20 

Data Sharing with
Harvard Dataverse

 In-person

 Virtual

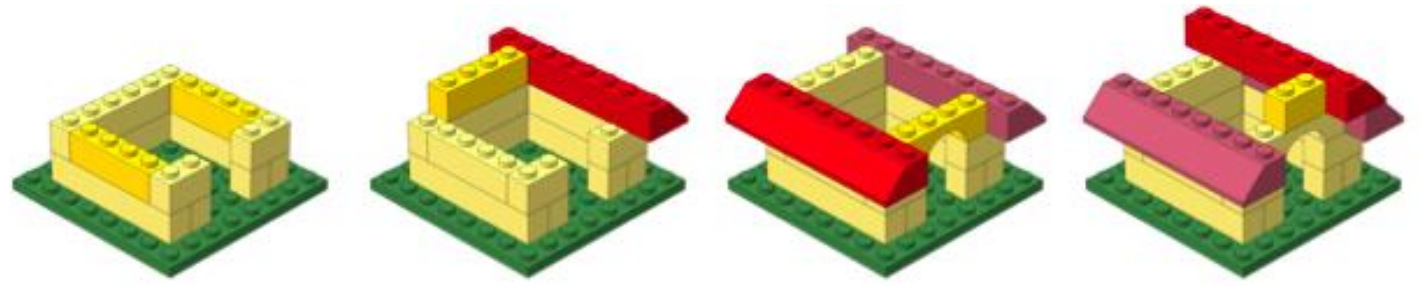


Learn More & Register
bit.ly/rdmwg-calendar



<https://datamanagement.hms.harvard.edu/training-events/rdmwg-calendar>

Keep building!



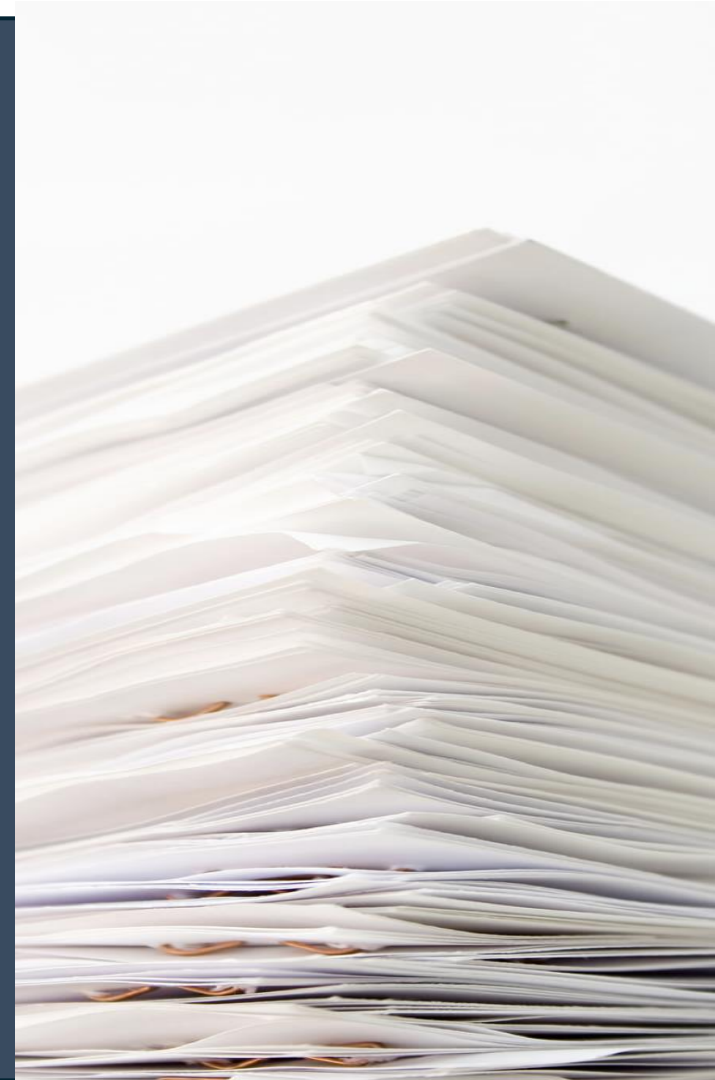
Workshop	Dates	Time	Location	Registration Page*
Introduction to Python	July 7, 10, 14 & 17 2026	9:30am-12pm	In-person	Register here
Introduction to R	August 28, September 1, 4 & 8 2026	10am-12pm	Zoom	Register here
Introduction to Python	November 10, 13, 17 & 20 2026	9:30am-12pm	Zoom	TBA

Workshop	Pre-requisite*	Dates	Time	Location	Registration Page**
Introduction to scRNA-seq (Python-based)	Introduction to Python	August 11, 14, 18 & 21, 2026	10am-12pm	Zoom	TBA
Introduction to Spatial Transcriptomics	Introduction to R	October 6, 9, 13 & 16, 2026	9:30am-12pm	Zoom	TBA
Tools for Reproducible Research	Introduction to R	October 27, 30 & November 3, 2026	9:30am-12pm	Zoom	Register here

<https://hsph.harvard.edu/research/bioinformatics/training/training-schedule/>

Talk to us early!

Involvement in study design to optimize experiments



More Information

- ❖ *HBC training materials: <https://hbctraining.github.io/main>*
- ❖ *HBC website: <https://hsph.harvard.edu/research/bioinformatics/>*

Contact Us

Sign up for our mailing list:

<https://tinyurl.com/hbc-training-mailing-list>

- ❖ *HBC training team:* hbctraining@hsph.harvard.edu
- ❖ *HBC consulting:* bioinformatics@hsph.harvard.edu